

14/07/25

Study of Various Network

Ex:) Commands in Windows and Linux

Aim:

Study of Various Networking commands used in linux and windows.

Basic Networking commands:

1) arp -a

Interface : 172.16.75.15 --- 0xb

Internet Address

Physical address Type

172.16.72.1

7C-5A-1C-6E-be-41 dynamic

172.16.72.60

04-f9-33-6C-37-91 dynamic

2) Hostname

DESKTOP-HSADBUG

3) Ipconfig /all

Windows IP configuration

Host name DESKTOP-HSADBUG

Primary DNS Suffix

Node Type Hybrid

4) Nbtstat -a

Display Protocol Statistics and current TCP/IP connection using NBT (NetBIOS over TCP/IP).

5) netstat

Active connections

Proto
TCP

Local address
127.0.0.1:49672

Foreign address
DESKTOP-HSADBUG

6) nslookup
Default Server: unknown
Address: 172.16.72.1

7) PathPing

Usage: PathPing [-g host-list] [-n maximum hops]
[-i address] [-n] [-P Period] [-g num - queries]
[-w timeout]

8) Route

Command is used to show or manipulate the IP routing table. It is primarily used to set up static route to specific host or network via an interface.

Linux:

1. ip:-
Command is one of the basic commands every administration will need in daily work.

ip [OPTIONS] <OBJECT> [COMMAND]

a) [root@Server ~] # ip address show

b) [root@Server ~] # ip address add 192.168.

c) [root@Server ~] # ip address add 192.168.1.254

d) [root@Server ~] # ip link set ens3 eth0 down

e) [root@Server ~] # ip link eth0 up

f) [root@Server ~] # ip link eth0 promisc on

g) [root@Server ~] # ip route add default via

h) [root@Server ~] # ip route add 192.168.1.254 via 192.168.1.1

i) [root@Server ~] # ip route get 10.10.1.4

2. ifconfig

command was is a staple in many Sysadmin's tool chest for configuring and trouble shooting network. It has since been

3. nmap:

Is a program with command-line interface that source is a network diagnostic and trouble shooting tool.

Syntax of command:

nmap [options] <hostname / IP>

a. [root@Server ~] # nmap google.com

b. [root@Server ~] # nmap google.com

c. [root@Server ~] # nmap google.com

d. [root@Server ~] # -c 10 google.com

4. tcpdump

The tcpdump command is designed for capturing and displaying packets.

a) [root@Server ~] # install -y tcpdump

b) [root@Server ~] # tcpdump -i

c) [root@Server ~] # tcpdump -i eth0

d) [root@Server ~] # tcpdump -i eth0 src host 8.8.8.8

e) [root@Server ~] # tcpdump -i eth0 net 10.0.0.0

f) [root@Server ~] # tcpdump -i eth0 net 10.0.0.0

g) [root@Server ~] # tcpdump -i eth0

h) [root@Server ~] # tcpdump -i eth0

2) [root@Server ~]# tcpdump -i eth0 port not 83

5. Ping

Tools that write IP level connectivity to another TCP/IP computer by sending Internet connect message Protocol. Echo Request message.

a) [root@Server ~]# ping google.com

b) [root@Server ~]# Ping -c 10 google.com

While Pinging about, you'll find different O/P from the from the Ping results

1) Destination host unreachable.

output:

i) ip link set ens3 up

ip link set ens3 down

ip link set ens3 Promise On

ip route add default via 192.168.1.254

ip route add 192.168.1.0/24 via 192.168.1.254

ip route delete 192.168.1.0/24 via 192.168.1.254

ip route get 10.10.1.4

2) ens3 flag = 10992 UP, BROADCAST, MULTICAST
ether 20:88:10:86:64 txqueuelen 1000

rx bytes 0 tx bytes 0

tx errors 0 rx errors 0 overruns 0 frame

flag = 73 UP, RUNNING, RUNNING

3) my tracert [v0.9.5]

fedora[192.16.79.86] => google.com(142.281.21.200)

2025-07-14 09:04:22 +0530

Keys: Help display mode Restart statistics order

field	Packet	Pings
Host	Cost %	src last Avg Best worst
1. gateway	0.61%	019 22 5.6 1.8 825
2. 142.280.171	0.01%	201 7.2 12.0 4.6 2532
3. 142.251.227	17.0%	261 175.5 185.0 1549 3442

b) mtr -g google.com

c) mtr -b google.com

d) mtr -c 10. google.com

4) O/P

a) dropped Privs to tcpdump

tcpdump: verbose output suppressed, use -V for full Protocol decode.

b) dropped Privs to tcpdump

tcpdump: verbose output suppressed, use -V for full Protocol decode.

Snapshot length 262244 bytes

o Packets captured

o Packets received by filter

o Packets dropped by filter.

3) dropped ~~Prins~~ to tcpdump
tcpdump . Verbose output Suppressed using
for full Protocol decode.

- o Packets Captured
- o Packets received by filter

5) O/P:

PING google.com (142.253.221.288) 56 (34)
bytes of data
from fedora (192.168.8.294) : icmp_seq=1
Destination host unreachable.

Result:

Various network commands used in
linux and windows has been executed
Successfully.

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